

Aircraft Design

Urban Air Mobility

Group 2 Final Presentation
March 31, 2025

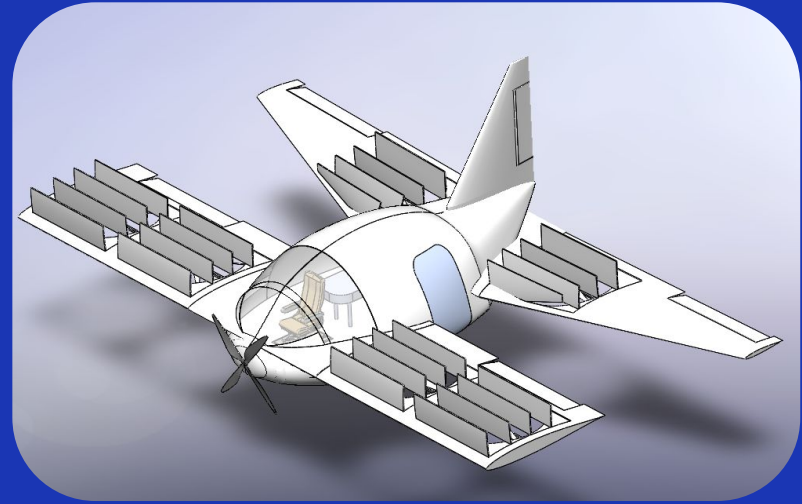


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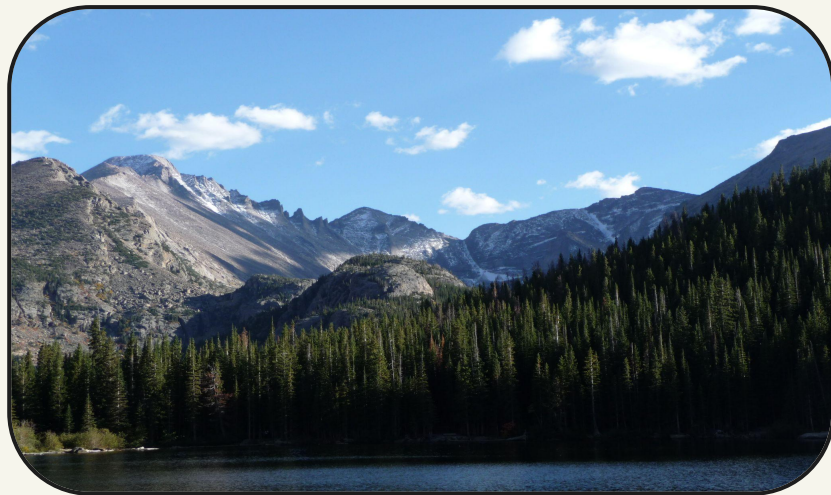
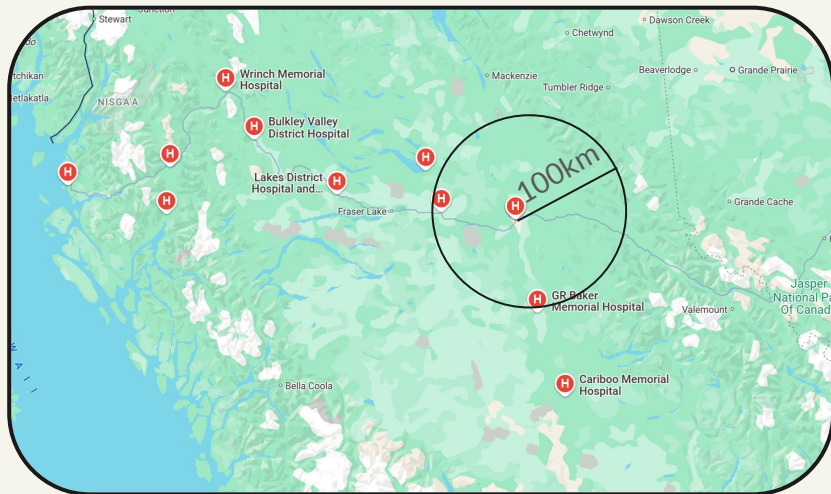
01 Background



Objectives & Problem Statement

To extend the rapid response capabilities of urban paramedics to remote areas in northern BC

- Emergency response
- Access remote areas
- Based out of Prince George Hospital
- Ability to carry emergency medical response personnel and equipment
- High speed during cruise allowing fast response times



Market Study



City Ambulance

Pros:

Medium Capacity
Close Range
Very Maneuverable

Cons:

Low Speed
Traffic Dependent



Air Ambulance

Pros:

Medium Capacity
High Speed
Moderate Range

Cons:

Moderate Maneuverability
Slightly Specialized



CH-149 Cormorant

Pros:

Large Capacity
High Speed
Long Range

Cons:

Low Maneuverability
Heavily Specialized
Very Expensive



Boeing V-22 Osprey

Pros:

Very Large Capacity
Very High Speed
Very Long Range

Cons:

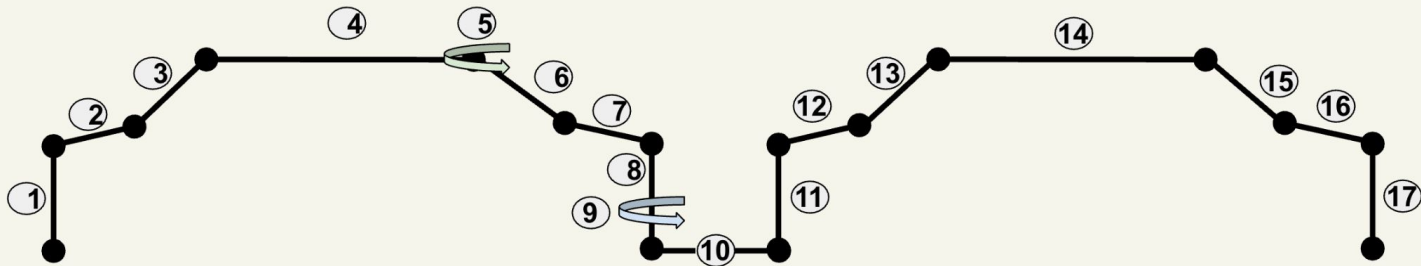
Very Large
Very Low Maneuverability
Very Expensive
Military Aircraft

Mission Profile

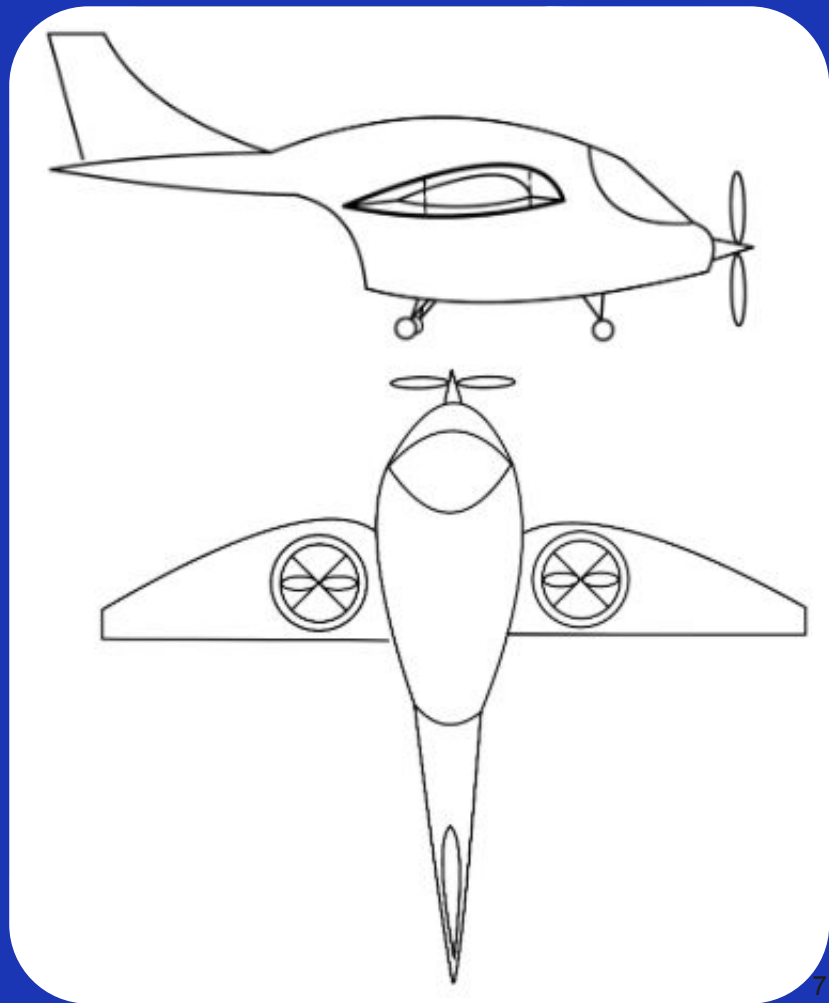
Our aircraft will be designed to travel up to 100 KM one way, for a total range of 200 KM

Stage	Description	Approx. Altitude
1	Vertical Climb	0-300m
2	Transition	300 - 310m
3	Forward Climb	310 - 1500m
4	Cruise	1500m
5	Loiter	1500m
6	Descent	1500 - 310m
7	Transition	310 - 300m
8	Vertical Descent	300 - 0m
9	Hover	10 - 1m
10	Pick-up	0m

11	Vertical Climb	0 - 300m
12	Transition	300 - 310m
13	Forward Climb	310 - 1500m
14	Cruise	1500m
15	Descent	1500 - 310m
16	Transition	310 - 300m
17	Vertical Descent	300 - 0m

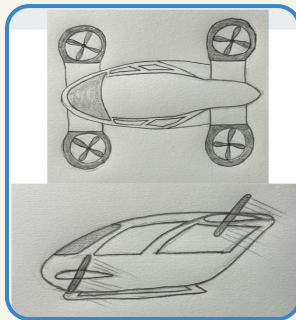


02 Concept Selection

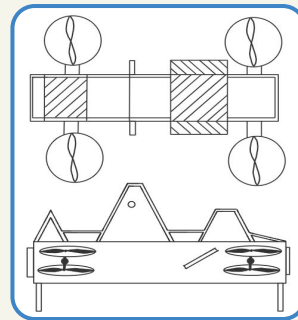


Concept Generation

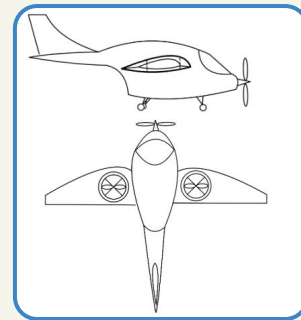
- Tilt Rotor Quadcopter
 - Hybrid Propulsion
 - Tilt rotor
- Fixed Rotor Quadcopter
 - Full electric
- Lift + Cruise with Ducted Fans
 - Hybrid
 - Separate forward and vertical propulsion
- Gyroplane
 - Hybrid
 - Powered rotor off in forward flight
- Tilt Rotor Micro-Jet
 - Wingtip motors
 - Fixed wing
- Dual Prop Lift + Cruise
 - Hybrid
 - Adaptive Robotic Landing Gear



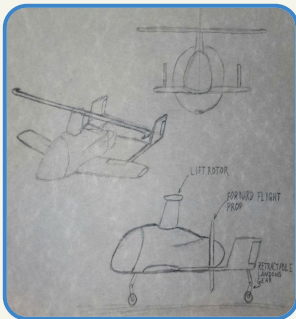
**Tilt Rotor
Quadcopter**



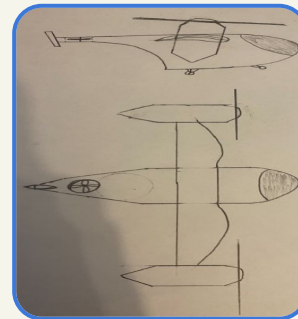
**Fixed Rotor
Quadcopter**



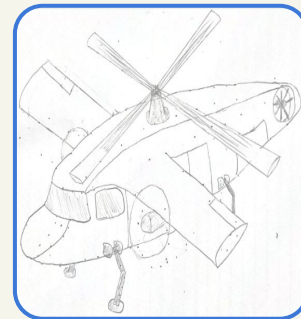
**Lift + Cruise with
Ducted Fans**



Gyroplane



Tilt Rotor Micro-Jet



**Dual Prop Lift +
Cruise**

Analytic Hierarchy Process

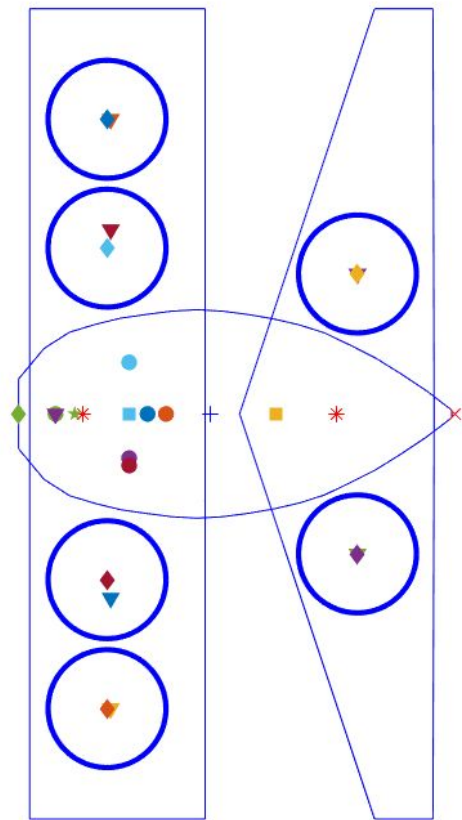
Criteria	Speed	Range	Pollution	Capacity	Total Area	Noise	Cost
Speed	1	7/1	7/1	3/1	3/1	9/1	8/1
Range	1/7	1	3/1	3/1	3/1	9/1	5/1
Pollution	1/7	1/3	1	1/6	1/8	1	1/6
Capacity	1/3	1/3	6/1	1	1	7/1	1/2
Total Area	1/3	1/3	8/1	1	1	7/1	2/1
Noise	1/9	1/9	1	1/7	1/7	1	1/6
Cost	1/8	1/5	6/1	2/1	1/2	6/1	1

Order of Importance:

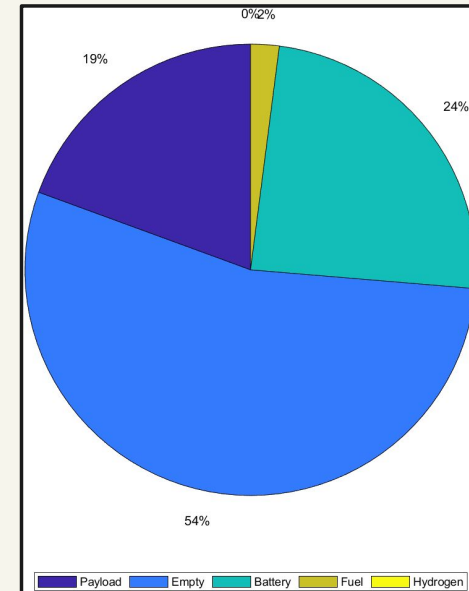
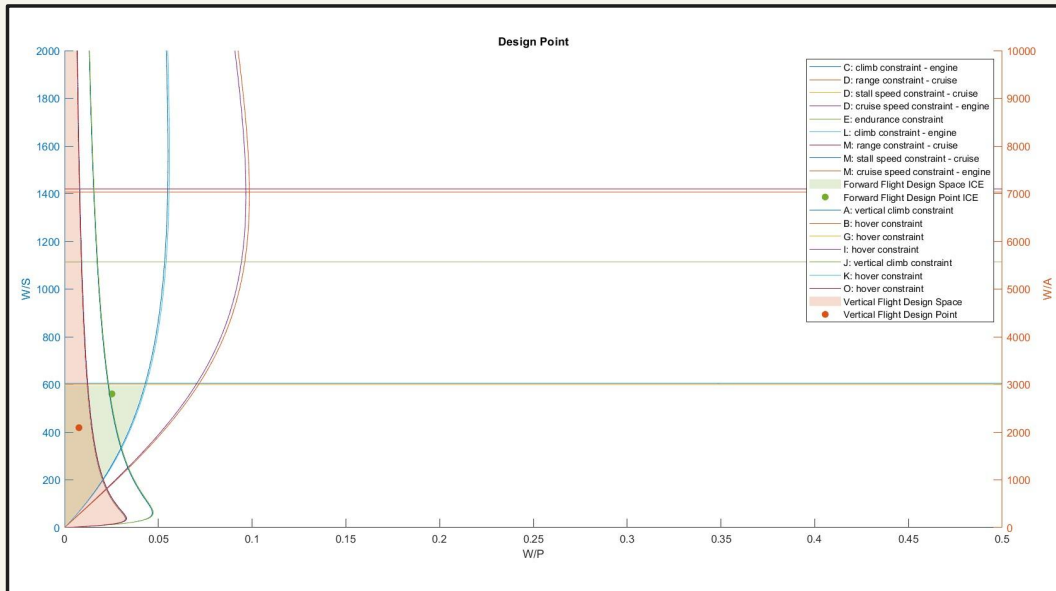
1. Speed
2. Plane area
3. Capacity
4. Noise
5. Cost
6. Range
7. Pollution

Winning Design Concept: Lift + Cruise with Ducted Fans

03 Design



MTOM & Design Point

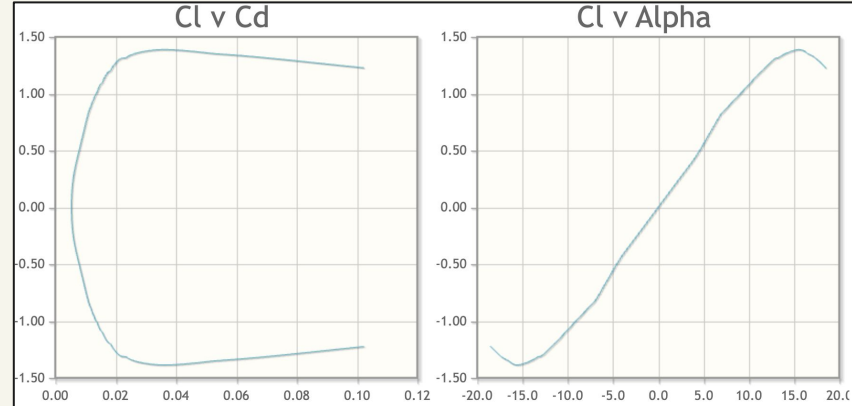


Final iterated MTOM of 2400 kg

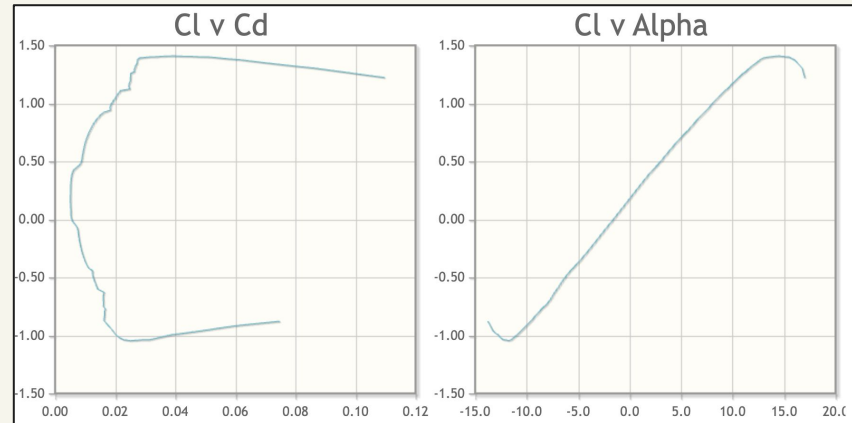
Main Wing & Tail Design

	Main Wing	Tandem Wing	Vertical Tail
Airfoil Type	NACA 63-210	NACA 63-210	NACA 0012
Taper Ratio	0	0.3	0.5
Aspect Ratio	4.62	6.41	1.5
Span	11 m	11 m	1.95 m
Sweep Angle	0	18.43°	35.0°

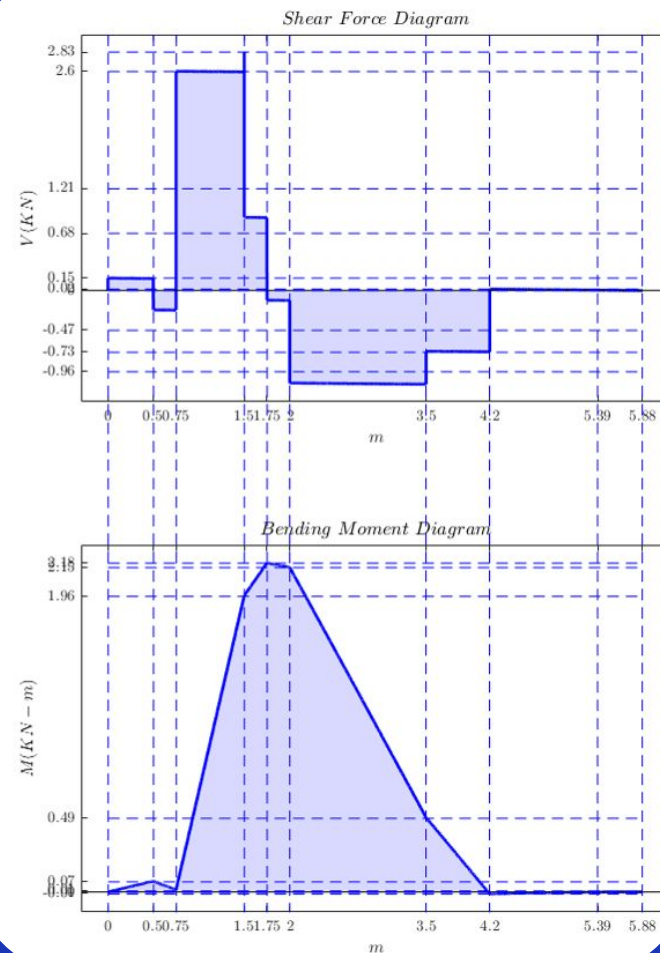
NACA 0012



NACA 63-210



04 Technical Data



Propulsion

Rotors & Fans

- Number of Fans: 6
- Blades per Fan: 5
- Rotor Diameter: 1.6 m
- Rotor Efficiency: 85%

Electric Motors

- Max Motor Power: 340 kW
- Required Motor Power: 170kW
- Motor Efficiency: 95%



Vertical Flight Conditions

- Climb Height: 0-300 m
- Climb Speed: 4m/s
- Total Required Time: 520s
- Design Time: 1040s

Battery Specifications

- Energy Density: 0.45 kWh/kg
- Capacity: 290 kWh
- Total Weight: 700 kg



Horizontal Flight Conditions

- Cruise Distance: 200 km
- Cruise Speed: 60 m/s
- Cruise Time: 1 hour

PBS TP100 Turboprop

- Max Power: 180 kW
- Cruise Power: 106 kW
- Weight: 62 kg



Jet A-1 Fuel

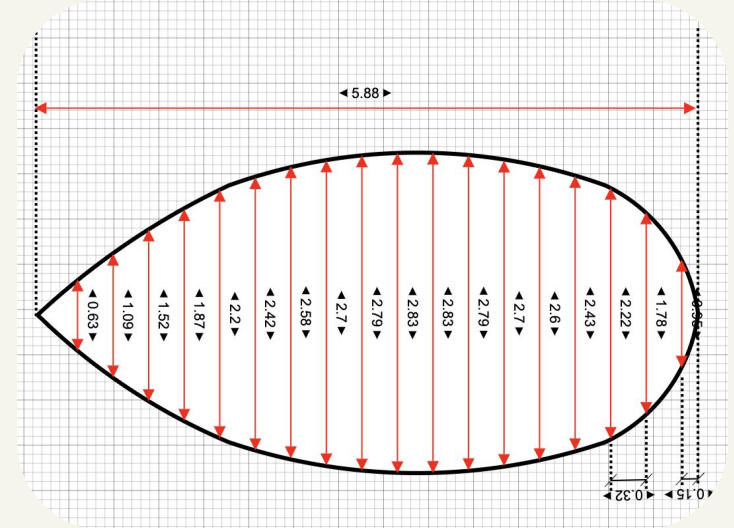
- Total Mass: 60 kg
- Reserve: 20%

Propeller

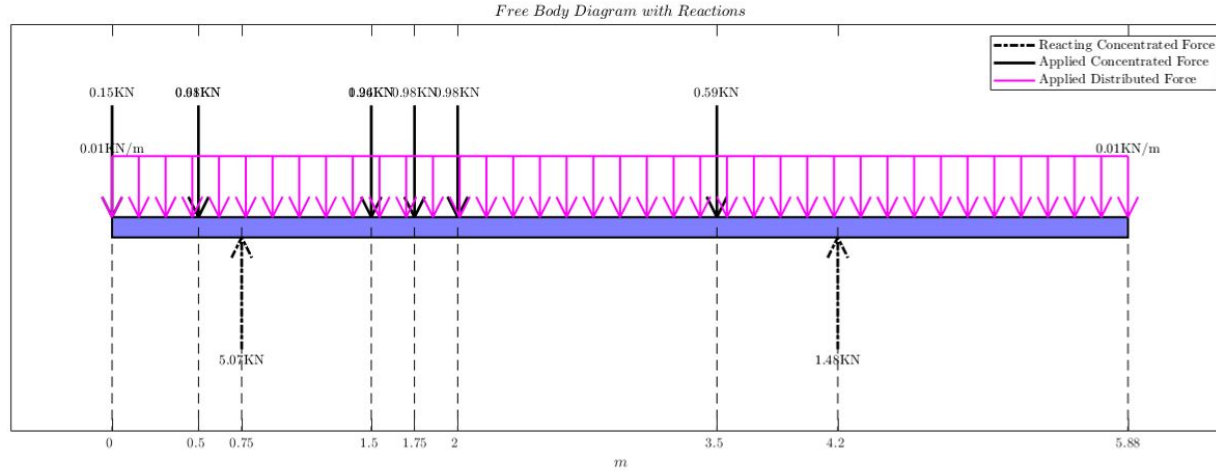
- Baldes: 4
- Diameter : 2m
- Efficiency: 87%



- **Fuel Tank Volume:** 0.069m^3
- **Base Drag Estimation:** $C_{D0} = 0.011$



Structural Design (Fuselage)

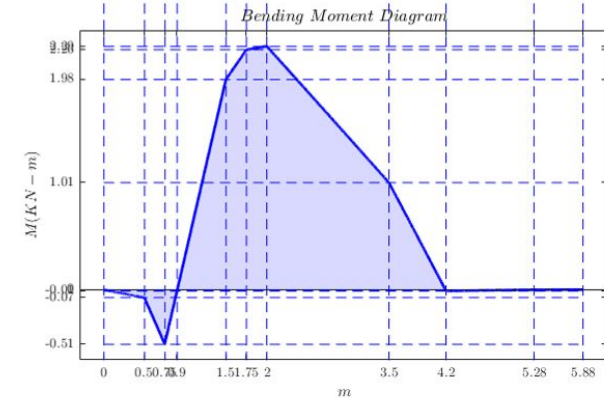
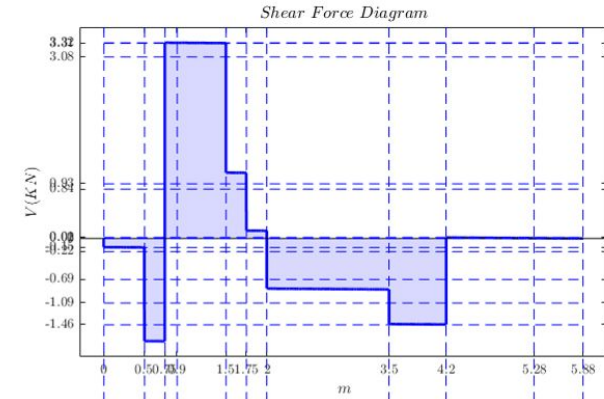


Range
 0 to 0.5
 0.5 to 0.75
 0.75 to 1.5
 1.5 to 1.75
 1.75 to 2
 2 to 3.5
 3.5 to 4.2
 4.2 to 5.88

Equations of Shear Force :

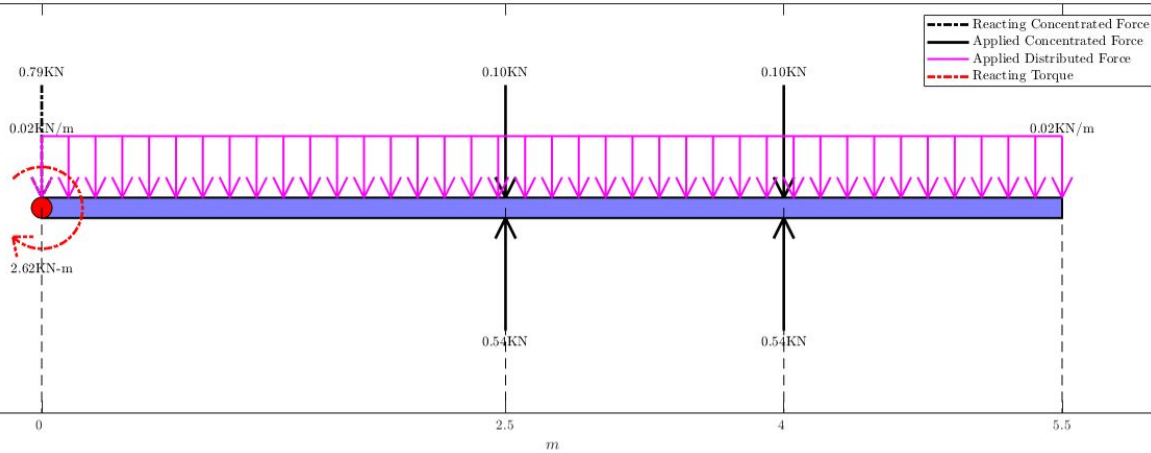
$$\begin{aligned}
 &-0.01x - 0.147 \\
 &-0.01x - 1.736 \\
 &-0.01x + 3.33 \\
 &-0.01x + 1.132 \\
 &-0.01x + 0.151 \\
 &-0.01x - 0.83 \\
 &-0.01x - 1.418 \\
 &-0.01x + 0.059
 \end{aligned}$$

Equations of Bending Moment :

$$\begin{aligned}
 &-0.005x^2 - 0.147x \\
 &-0.005x^2 - 1.736x + 0.795 \\
 &-0.005x^2 + 3.33x - 3.005 \\
 &-0.005x^2 + 1.132x + 0.292 \\
 &-0.005x^2 + 0.151x + 2.009 \\
 &-0.005x^2 - 0.83x + 3.971 \\
 &-0.005x^2 - 1.418x + 6.029 \\
 &-0.005x^2 + 0.059x - 0.171
 \end{aligned}$$


Structural Design (Main Wing Take-off)

Free Body Diagram with Reactions



Range
0 to 2.5
2.5 to 4
4 to 5.5

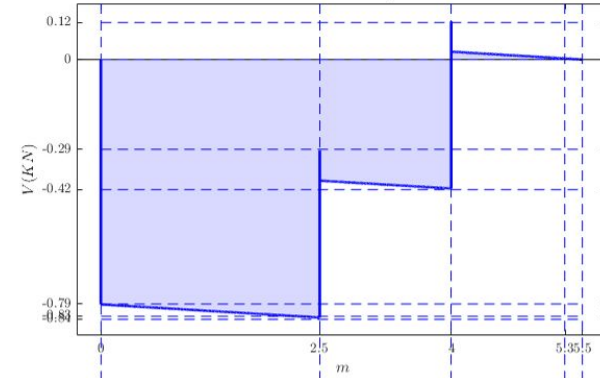
Equations of Shear Force :

$-0.017x - 0.792$
 $-0.017x - 0.349$
 $-0.017x + 0.094$

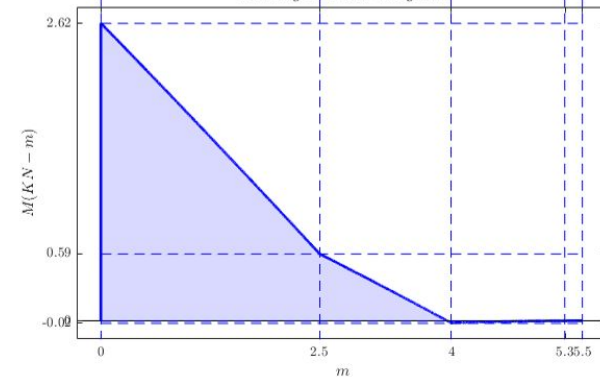
Equations of Bending Moment :

$-0.009x^2 - 0.792x + 2.622$
 $-0.009x^2 - 0.349x + 1.515$
 $-0.008x^2 + 0.093x - 0.257$

Shear Force Diagram

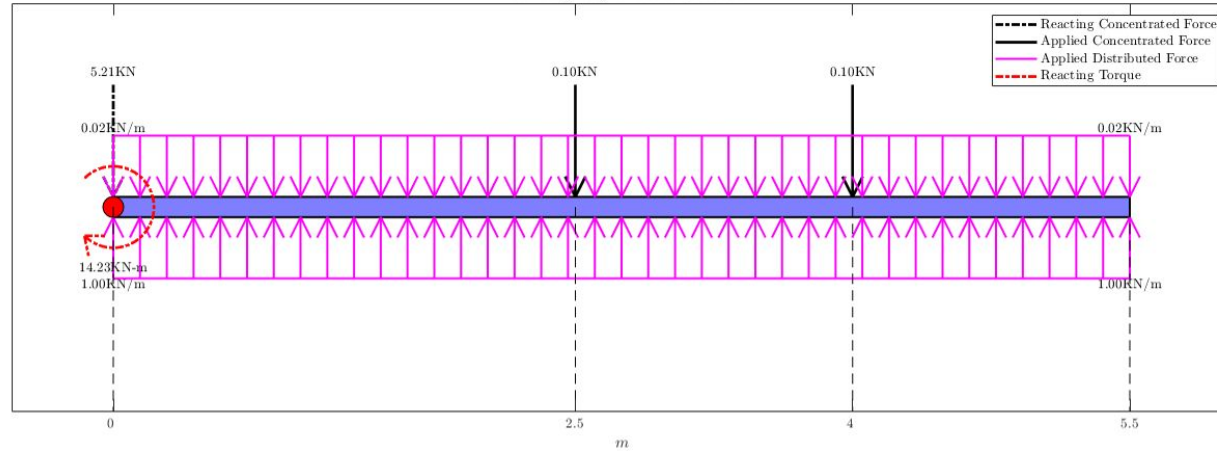


Bending Moment Diagram



Structural Design (Main Wing Cruise)

Free Body Diagram with Reactions



Range
0 to 2.5
2.5 to 4
4 to 5.5

Equations of Shear Force :

$$0.983x - 5.21$$

$$0.983x - 5.308$$

$$0.983x - 5.407$$

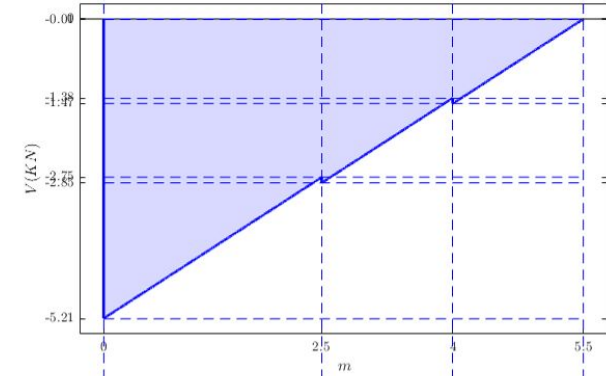
Equations of Bending Moment :

$$0.491x^2 - 5.21x + 14.23$$

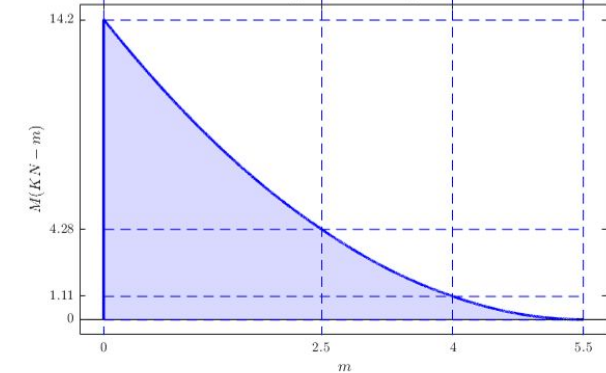
$$0.491x^2 - 5.308x + 14.475$$

$$0.492x^2 - 5.407x + 14.868$$

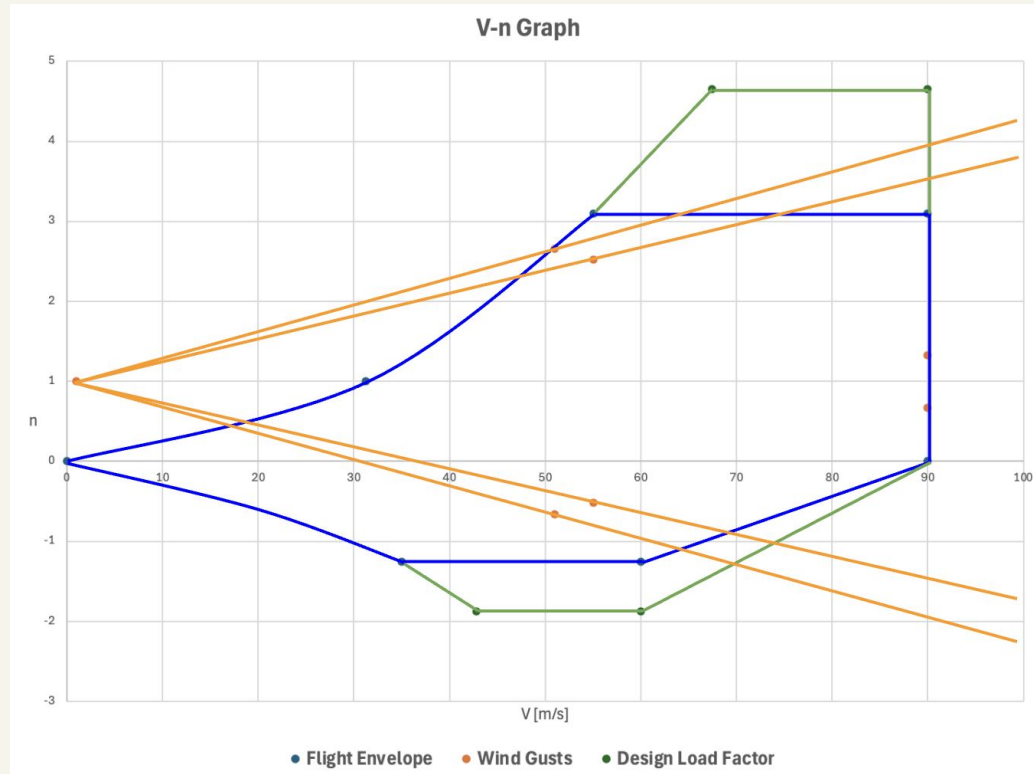
Shear Force Diagram



Bending Moment Diagram



Structural Design



Static Stability

Position of the forces on the schematic:

$$X_{C.G.} = 0.45 \text{ m}$$

$$X_{N.P.} = 1.143 \text{ m}$$

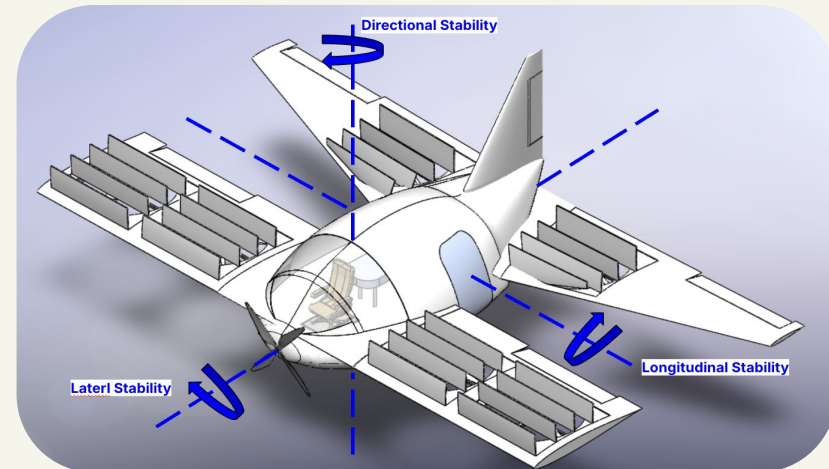
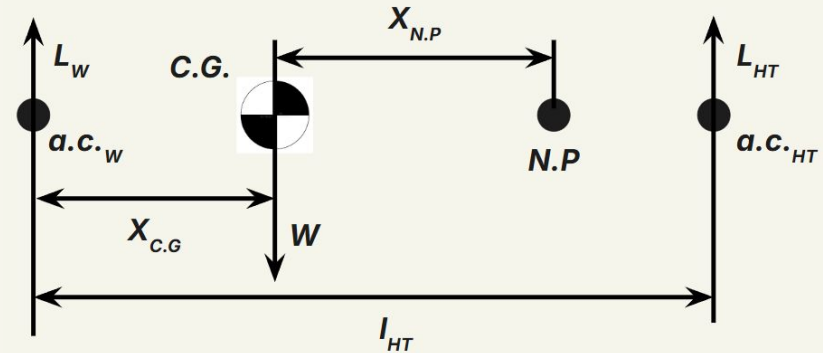
$$l_{HT} = 2.54 \text{ m}$$

Aircraft is Statically Stable:

Coefficient of Longitudinal Stability, $C_{M\alpha} = -1.47$

Coefficient of Lateral Stability, $C_{L\beta} = -0.0825$

Coefficient of Directional Stability, $C_{n\beta} = 0.0825$



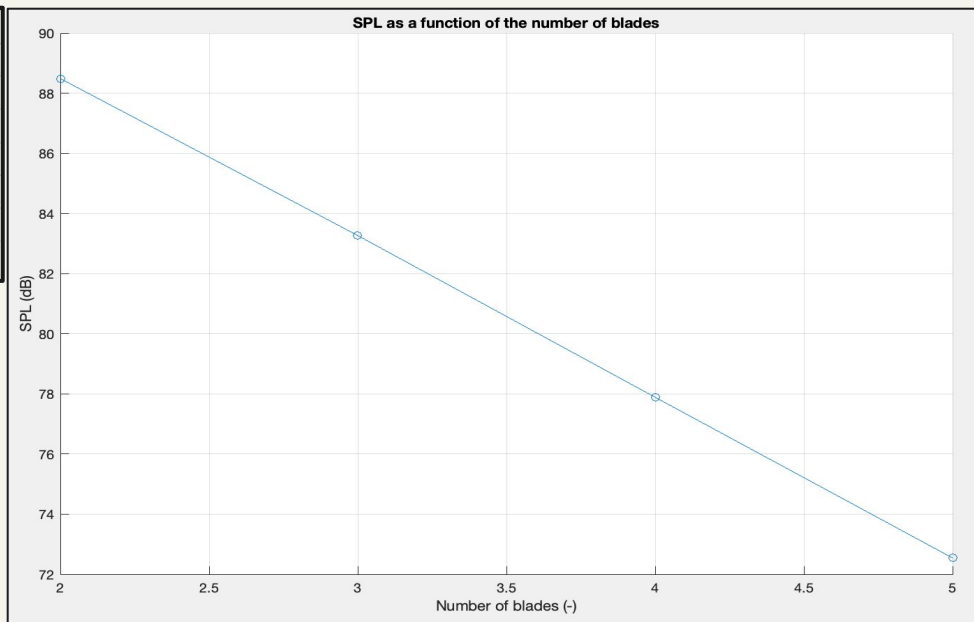
Emissions

Pollution:

Jet Fuel A1	Emissions [kg CO2] per refill		
capacity	Production	burnt	total
60kg	226.8	189.6	416.4
Li-ion Battery	Emissions [kg CO2] per cycle		
capacity	Production	per-charge	total
290kW.h	12.702	2.871	15.573
Battery Service Life	15 Years	~2000cycles	

Noise:

Variable	Value
Harmonic Number	1
Number of Blades	5
Number of Rotors	6
Radius (m)	0.8
Solidity Ratio (%)	8
Max t/c	0.12
Mach Tip	0.755
Power Per Rotor (kW)	120
Thrust Per Rotor (N)	261.25
Altitude from Observer (ft)	150



Max SPL = 72.5 dB from 150 ft away
 (Equivalent to a washing machine or vacuum cleaner)

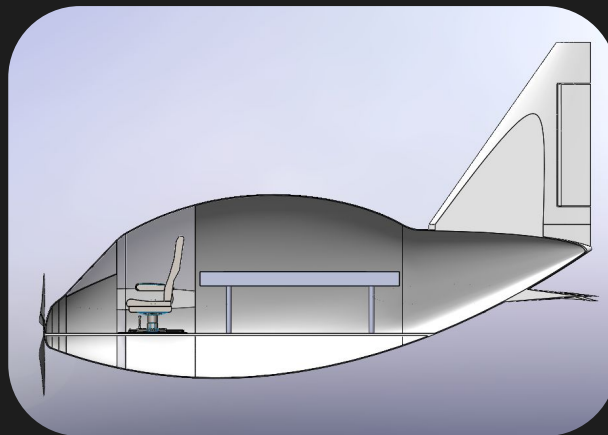
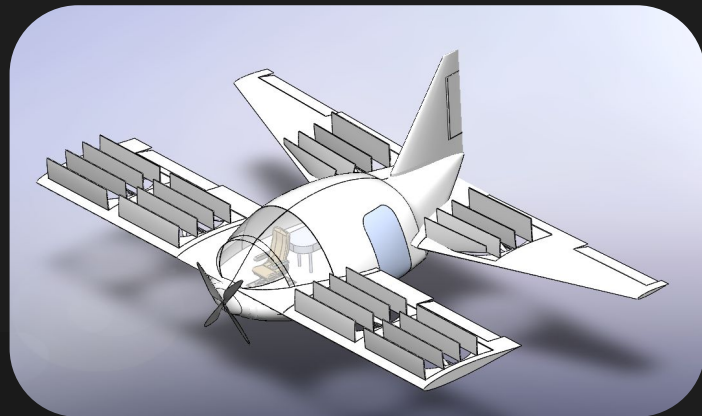
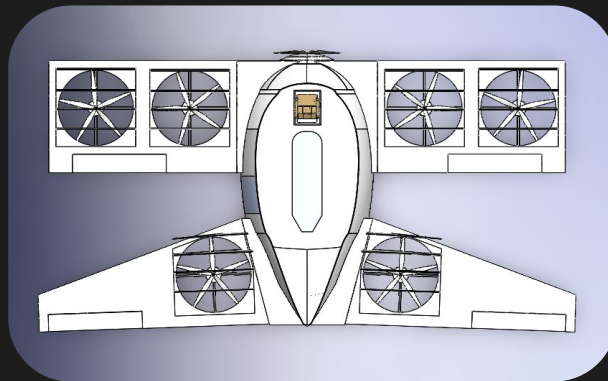
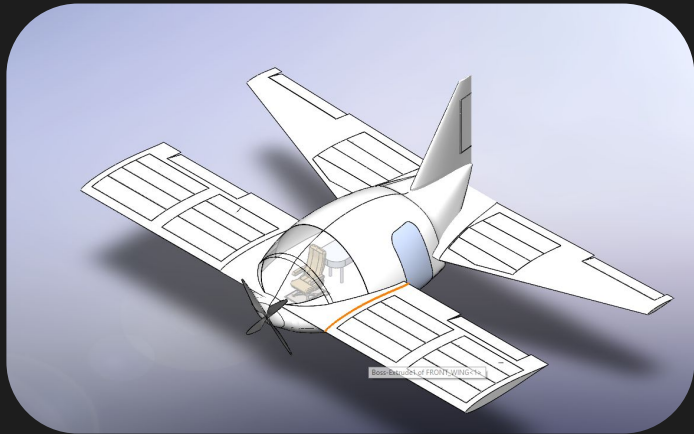
Cost

- CPI forecasted to increase by ~2%/yr
- Profit margin of 5.9%, or 0.25M\$ per unit
- Qp of 50 to serve BC hospitals

UNIT COST	M\$
Emergency Aircraft	5.345
UNIT PRICE	
Emergency Aircraft	5.595

Variable	Value	Units
AEM	2400	kg
Vmax	60	m/s
Qd	10	--
Qp	50	--
CPI	2	--
Engine type (K)	520	--
Max Static Thrust	603.45	lb-f
Unit Profit	0.25	M\$
Max Power	180000	W
Number engines	1	--

CAD Model



Thanks!

Questions?